4

a) The URL is gaia.cs.umass.edu/cs453/index.html

b) The browser is running HTML 1.1

c) The browser requests a persistent connection, because of “Connection: keep-alive“

d) The host’s IP is gaia.cs.umass.edu

e) The bowser type that initiates the message is mozilla. The browser type is needed because different browsers process web pages differently

5

a) The server was able to successfully find the document and returned response code 200. The date was Tue, 07 Mar 2008 12:39:45 GMT

b) Last-Modified: Sat, 10 Dec 2005 18:27:46 GMT

c) Content-Length: 3874

d) The first 5 bytes of the document being returned are , and the server agreed to a persistent connection

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DNS servers = n

RTT = RTT1, … , RTTn

objs = 1

Trans time = 0s

how much time elapses from when the client clicks on the link until the client receives the object?

= base (2 RTT) + visits (RTT1 + … + RTTn) + obj (1 RTT)

8

a) 2 RTT per item = 2 * 8 = 16

b)

c) Total = 1 RTT + # objs = 1 + 8 = 9 RTT

10

Note: assume all the transmissions on the parallel connections are 100% parallel

10meter link:

Trans = 150 bits/sec in both directions (R)

Packets 100,000 bits (L)

packets containing only control (e.g., ACK or handshaking) are 200 bits long (L?)

N parallel connections

each get 1/N of the link bandwidth.

HTTP protocol:

each downloaded object is 100 Kbits long

initial downloaded object contains 10 referenced objects from the same sender

Would parallel downloads via parallel instances of non-persistent HTTP make sense in this case?

dhop = dproc + dqueue + dtrans + dprop

= (100000/150) + (200/150)

Now consider persistent HTTP. Do you expect significant gains over the non-persistent case

11

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Take snapshots of cache and record the server that‘s appears the most